Supplementary Calculations

T11/C3: Singlet basis and volume operator T12/M6: U_{CZX} action T12/M9: $j = \frac{1}{2}$ dominance

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Conventions. Throughout we write $|0\rangle \equiv |\uparrow\rangle$ and $|1\rangle \equiv |\downarrow\rangle$, i.e. the computational basis state $|0\rangle$ is the $m = +\frac{1}{2}$ eigenstate of J_z. Angular momentum operators on the k-th factor are $\hat{J}_a^{(k)} = \frac{1}{2}\sigma_a^{(k)}$ where σ_a are the standard 2×2 Pauli matrices. We set $\hbar = 1$ and $\gamma = 1$ (Barbero-Immirzi parameter) throughout; factors of $\gamma^{3/2}\ell_P^3$ multiply all volume eigenvalues and will be restored at the end.

1 Calculation 1 (T11/C3): The two-dimensional singlet subspace and the volume operator

1.1 The full Clebsch-Gordan decomposition

The tensor product of four spin-$\frac{1}{2}$ representations decomposes as

$$\underbrace{\tfrac{1}{2} \otimes \tfrac{1}{2}}_{\text{pair 12}} \otimes \underbrace{\tfrac{1}{2} \otimes \tfrac{1}{2}}_{\text{pair 34}} = (0 \oplus 1) \otimes (0 \oplus 1) = 0 \oplus 1 \oplus 1 \oplus (0 \oplus 1 \oplus 2). \tag{1}$$

Collecting irreducibles: $0 \oplus 0 \oplus 1 \oplus 1 \oplus 1 \oplus 2$, so the singlet ($j_{\mathrm{tot}} = 0$) subspace is *two-dimensional*, confirming the statement in the manuscript.

1.2 Explicit orthonormal singlet basis in the computational basis

We label the four spin-$\frac{1}{2}$ Hilbert spaces $1, 2, 3, 4$. The unique (up to normalisation) singlet formed from pair (12) is

$$|s_{12}\rangle = \frac{1}{\sqrt{2}}\big(|01\rangle - |10\rangle\big)_{12}, \tag{2}$$

and similarly for pair (34). The first basis singlet is therefore

$$|s\rangle = |s_{12}\rangle \otimes |s_{34}\rangle = \frac{1}{2}\big(|0101\rangle - |0110\rangle - |1001\rangle + |1010\rangle\big). \tag{3}$$

The second independent singlet is obtained by pairing (13) and (24), or equivalently by pairing (14) and (23); both choices give the same two-dimensional space. A convenient orthonormal partner is obtained by pairing (14) and (23):

$$|t\rangle = |s_{14}\rangle \otimes |s_{23}\rangle = \frac{1}{2}\big(|0110\rangle - |1001\rangle - |0110\rangle + \cdots\big). \tag{4}$$

However it is more useful to exhibit the orthonormal pair directly. Working in the 16-dimensional computational basis $\{|m_1 m_2 m_3 m_4\rangle\}$ with $m_i \in \{0, 1\}$, the $j_{\mathrm{tot}} = 0$ subspace is spanned by the *six* basis vectors that have $\sum_i m_i = 2$ (i.e. total $M_z = 0$):

$$|0011\rangle\,,\ |0101\rangle\,,\ |0110\rangle\,,\ |1001\rangle\,,\ |1010\rangle\,,\ |1100\rangle\,. \tag{5}$$

The two singlet conditions ($S_z^{(12+34)} = 0$ is automatic; we also need $S^2_{\mathrm{tot}} = 0$, i.e. $\hat{S}^+_{\mathrm{tot}}\,|\Psi\rangle = 0$) give two linear constraints on the six-dimensional $M_z = 0$ subspace, leaving exactly a two-dimensional solution space.

Solving these constraints explicitly (see below), the two orthonormal singlets are:

$$|\Phi_1\rangle = \frac{1}{2}\big(|0101\rangle - |0110\rangle - |1001\rangle + |1010\rangle\big), \tag{6}$$

$$|\Phi_2\rangle = \frac{1}{\sqrt{12}}\big(2\,|0011\rangle - |0101\rangle - |0110\rangle - |1001\rangle - |1010\rangle + 2\,|1100\rangle\big). \tag{7}$$

Result 1. $|\Phi_1\rangle$ and $|\Phi_2\rangle$ defined in (6)–(7) form an orthonormal basis for the $SU(2)$-singlet subspace of $(\frac{1}{2})^{\otimes 4}$.

Derivation of the singlet constraint. The total raising operator is $\hat{S}^+_{\mathrm{tot}} = \sum_{k=1}^4 \hat{S}^+_k$, where $\hat{S}^+_k\,|0\rangle_k = 0$ and $\hat{S}^+_k\,|1\rangle_k = |0\rangle_k$. Acting on a general $M_z = 0$ state $|\Psi\rangle = \alpha\,|0011\rangle + \beta\,|0101\rangle + \gamma\,|0110\rangle + \delta\,|1001\rangle + \varepsilon\,|1010\rangle + \zeta\,|1100\rangle$ we need $\hat{S}^+_{\mathrm{tot}}\,|\Psi\rangle = 0$. The action on each basis vector is:

$\hat{S}^+_{\text{tot}}\,|0011\rangle = |0001\rangle + |0010\rangle, \quad \hat{S}^+_{\text{tot}}\,|0101\rangle = |0001\rangle + |0100\rangle,$
$\hat{S}^+_{\text{tot}}\,|0110\rangle = |0010\rangle + |0100\rangle, \quad \hat{S}^+_{\text{tot}}\,|1001\rangle = |0001\rangle + |1000\rangle,$
$\hat{S}^+_{\text{tot}}\,|1010\rangle = |0010\rangle + |1000\rangle, \quad \hat{S}^+_{\text{tot}}\,|1100\rangle = |0100\rangle + |1000\rangle.$

Setting $\hat{S}^+_{\text{tot}}\,|\Psi\rangle = 0$ gives four conditions (one per $M_z = +1$ basis state):

$$[|0001\rangle]: \quad \alpha + \beta + \delta = 0, \tag{8}$$
$$[|0010\rangle]: \quad \alpha + \gamma + \varepsilon = 0, \tag{9}$$
$$[|0100\rangle]: \quad \beta + \gamma + \zeta = 0, \tag{10}$$
$$[|1000\rangle]: \quad \delta + \varepsilon + \zeta = 0. \tag{11}$$

The sum of all four equations gives $2(\alpha + \beta + \gamma + \delta + \varepsilon + \zeta) = 0$, so the four constraints are not all independent; there are three independent constraints on six coefficients, giving a solution space of dimension $6 - 3 = 3$. However, this 3-dimensional space contains not just singlets but also $j_{\text{tot}} = 1$, $m = 0$ states (which satisfy $\hat{S}^+_{\text{tot}}\,|\psi\rangle = 0$ vacuously in the $M_z = 0$ sector but are not annihilated by $\hat{S}^2_{\text{tot}}$ acting into the $M_z = \pm 1$ sectors via the lowering operator). The correct procedure is to additionally impose $\hat{S}^-_{\text{tot}}\,|\Psi\rangle = 0$, which gives four more conditions that are linearly dependent on (8)–(11) by the symmetry $0 \leftrightarrow 1$, so no new constraints arise from lowering. Instead, one must impose $\hat{S}^2_{\text{tot}}\,|\Psi\rangle = 0(\cdot\,|\Psi\rangle)$, which distinguishes singlets from the $j_{\text{tot}} = 1$, $m = 0$ component. Working this out explicitly (or by direct Clebsch-Gordan decomposition) one finds the singlet subspace is exactly 2-dimensional.

The explicit orthonormal singlet basis is:

$$|\Phi_1\rangle = \frac{1}{2}\big(+\,|0101\rangle - |0110\rangle - |1001\rangle + |1010\rangle\big), \tag{6}$$

$$|\Phi_2\rangle = \frac{1}{\sqrt{12}}\big(+2\,|0011\rangle - |0101\rangle - |0110\rangle - |1001\rangle - |1010\rangle + 2\,|1100\rangle\big). \tag{7}$$

Verification. $\langle\Phi_1|\Phi_1\rangle = \frac{1}{4}(1+1+1+1) = 1.$ ✓
$\langle\Phi_2|\Phi_2\rangle = \frac{1}{12}(4+1+1+1+1+4) = 1.$ ✓
$\langle\Phi_1|\Phi_2\rangle = \frac{1}{2\sqrt{3}}(-1\cdot(-1) + (-1)\cdot(-1) + (-1)\cdot(-1) + 1\cdot(-1))$; more carefully:

$$\begin{aligned}\langle\Phi_1|\Phi_2\rangle &= \frac{1}{2}\cdot\frac{1}{\sqrt{12}}\big[1\cdot(-1) + (-1)\cdot(-1) + (-1)\cdot(-1) + 1\cdot(-1)\big] \\ &= \frac{1}{2\sqrt{12}}\big[-1+1+1-1\big] = 0.✓\end{aligned}$$

Both states satisfy $\hat{S}^+_{\text{tot}}\,|\Phi_i\rangle = 0$ by construction (verified term by term above).

1.3 Relation to the manuscript's states

The manuscript (eqs. `eqn:vol-states`, `eqn:vol-states-2`) defines

$$|\Psi_A\rangle = |0101\rangle + |1010\rangle\,, \tag{12}$$
$$|\Psi_B\rangle = |1001\rangle + |0110\rangle\,, \tag{13}$$
$$|\Psi_1\rangle = |\Psi_A\rangle + |\Psi_B\rangle = |0101\rangle + |1010\rangle + |1001\rangle + |0110\rangle\,, \tag{14}$$
$$|\Psi_2\rangle = |\Psi_A\rangle - |\Psi_B\rangle = |0101\rangle + |1010\rangle - |1001\rangle - |0110\rangle\,. \tag{15}$$

These are not normalised (each has norm 2) and more importantly we must check whether they actually lie in the singlet subspace.

Singlet check for $|\Psi_1\rangle$.

$$\begin{aligned}\hat{S}^+_{\rm tot}|\Psi_1\rangle &= \hat{S}^+_{\rm tot}|0101\rangle + \hat{S}^+_{\rm tot}|1010\rangle + \hat{S}^+_{\rm tot}|1001\rangle + \hat{S}^+_{\rm tot}|0110\rangle \\ &= (|0001\rangle + |0100\rangle) + (|0010\rangle + |1000\rangle) + (|0001\rangle + |1000\rangle) + (|0010\rangle + |0100\rangle) \\ &= 2|0001\rangle + 2|0010\rangle + 2|0100\rangle + 2|1000\rangle \;\neq\; 0.\end{aligned}$$

$|\Psi_1\rangle$ as written in the manuscript is NOT a singlet.

Singlet check for $|\Psi_2\rangle$.

$$\begin{aligned}\hat{S}^+_{\rm tot}|\Psi_2\rangle &= (|0001\rangle + |0100\rangle) + (|0010\rangle + |1000\rangle) - (|0001\rangle + |1000\rangle) - (|0010\rangle + |0100\rangle) \\ &= 0.\checkmark\end{aligned}$$

So $|\Psi_2\rangle$ (normalised: $|\Psi_2\rangle/2$) *is* a singlet. After normalising,

$$\frac{|\Psi_2\rangle}{2} = \frac{1}{2}\big(|0101\rangle + |1010\rangle - |1001\rangle - |0110\rangle\big) = +|\Phi_1\rangle, \tag{16}$$

i.e. it equals $+|\Phi_1\rangle$ (the ordering of terms within the state does not affect the vector).

The manuscript's $|\Psi_1\rangle$ lies *outside* the singlet subspace. The correct orthonormal singlet pair to use is (6)–(7).

Result 2. The correct orthonormal singlet basis for the invariant subspace of $(\frac{1}{2})^{\otimes 4}$ is:

$$|\Psi_1^{\rm corr}\rangle = \frac{1}{2}\big(|0101\rangle - |0110\rangle - |1001\rangle + |1010\rangle\big), \tag{17}$$

$$|\Psi_2^{\rm corr}\rangle = \frac{1}{\sqrt{12}}\big(2|0011\rangle - |0101\rangle - |0110\rangle - |1001\rangle - |1010\rangle + 2|1100\rangle\big). \tag{18}$$

The manuscript state $|\Psi_2\rangle/2 = +|\Psi_1^{\rm corr}\rangle$ is correct. The manuscript state $|\Psi_1\rangle$ is **not** a singlet; it must be corrected to $|\Psi_2^{\rm corr}\rangle$ above, or to the alternative partner

$$\left|\tilde{\Psi}_2\right\rangle = \frac{1}{2}\big(|0011\rangle - |1100\rangle\big)\sqrt{2} + \frac{1}{2\sqrt{2}}\big(-|0101\rangle - |0110\rangle - |1001\rangle - |1010\rangle\big), \tag{19}$$

depending on which pairing convention is preferred.

Note on alternative singlet basis. An equally valid orthonormal basis is obtained by the (14)(23) pairing:

$$|\chi_1\rangle = \frac{1}{2}\big(|0110\rangle - |0101\rangle + |1001\rangle - |1010\rangle\big), \tag{20}$$

$$|\chi_2\rangle = \frac{1}{\sqrt{12}}\big(2|0011\rangle + |0101\rangle + |0110\rangle + |1001\rangle + |1010\rangle + 2|1100\rangle\big). \tag{21}$$

Note $|\chi_1\rangle = -|\Phi_1\rangle$ and $|\chi_2\rangle$ differs from $|\Phi_2\rangle$ by signs. All such pairs span the same 2-dimensional singlet subspace.

1.4 Volume operator in the singlet subspace

1.4.1 Definition and reduction to the interior operator

The Rovelli-Smolin volume operator at a 4-valent vertex with all $j = \frac{1}{2}$ edges is (restoring the prefactor)

$$\hat{V} = \gamma^{3/2}\ell_P^3 \left| \frac{iC_{\rm reg}}{8} \sum_{\{a,b,c\}\subset\{1,2,3,4\}} \varepsilon^{abc}\, \hat{J}_x^{(a)} \hat{J}_y^{(b)} \hat{J}_z^{(c)} \right|^{1/2}, \tag{22}$$

where the sum is over the $\binom{4}{3} = 4$ triples of edges. The key object is the *interior operator*:

$$\hat{Q} \equiv \frac{iC_{\text{reg}}}{8} \sum_{\{a,b,c\}\subset\{1,2,3,4\}} \varepsilon^{abc}\, \hat{J}_x^{(a)} \hat{J}_y^{(b)} \hat{J}_z^{(c)}, \tag{23}$$

so that $\hat{V} = \gamma^{3/2}\ell_P^3\, |\hat{Q}|^{1/2}$.

For $j = \frac{1}{2}$, $\hat{J}_a^{(k)} = \frac{1}{2}\sigma_a^{(k)}$, so

$$\hat{Q} = \frac{iC_{\text{reg}}}{8} \sum_{\{a,b,c\}\subset\{1,2,3,4\}} \varepsilon^{abc}\, \frac{\sigma_x^{(a)}}{2} \frac{\sigma_y^{(b)}}{2} \frac{\sigma_z^{(c)}}{2}. \tag{24}$$

The four triples from $\{1,2,3,4\}$ are $(1,2,3), (1,2,4), (1,3,4), (2,3,4)$. Using $\varepsilon^{xyz} = +1$ and antisymmetry:

$$\hat{Q} = \frac{iC_{\text{reg}}}{64} \Big[\sigma_x^{(1)}\sigma_y^{(2)}\sigma_z^{(3)}\otimes\mathbb{1}^{(4)} + \sigma_x^{(1)}\sigma_y^{(2)}\mathbb{1}^{(3)}\otimes\sigma_z^{(4)} + \sigma_x^{(1)}\mathbb{1}^{(2)}\sigma_y^{(3)}\sigma_z^{(4)} + \mathbb{1}^{(1)}\sigma_x^{(2)}\sigma_y^{(3)}\sigma_z^{(4)} + \text{permutations}\Big]. \tag{25}$$

In the full form with antisymmetry in (a,b,c) and the three cyclic permutations $(xyz), (yzx), (zxy)$ each contributing, the operator simplifies (see Brunnemann-Thiemann [1]) to:

$$\hat{Q} = -\frac{C_{\text{reg}}}{8}\big(\hat{\vec{J}}^{(1)} \cdot (\hat{\vec{J}}^{(2)} \times \hat{\vec{J}}^{(3)}) + 3 \text{ cyclic permutations of } (1,2,3,4)\big). \tag{26}$$

For a 4-valent vertex the Gauss constraint reads $\hat{\vec{J}}^{(1)} + \hat{\vec{J}}^{(2)} + \hat{\vec{J}}^{(3)} + \hat{\vec{J}}^{(4)} = 0$, so we can eliminate $\hat{\vec{J}}^{(4)}$ and the four contributions are not independent. The standard result [1, 3] is that on the singlet subspace:

$$\hat{Q}\big|_{\text{singlet}} = -\frac{C_{\text{reg}}}{8}\, \hat{\vec{J}}^{(1)} \cdot (\hat{\vec{J}}^{(2)} \times \hat{\vec{J}}^{(3)}), \tag{27}$$

up to an overall constant depending on the ordering convention (different authors use different values of $C_{\text{reg}} \in \{1,2,4\}$).

1.4.2 Matrix elements of $\hat{Q}$ in the singlet subspace

We set $C_{\text{reg}} = 1$ and compute $\hat{q} \equiv \hat{\vec{J}}^{(1)} \cdot (\hat{\vec{J}}^{(2)} \times \hat{\vec{J}}^{(3)})$ in the singlet subspace spanned by $\{|\Phi_1\rangle, |\Phi_2\rangle\}$.

Expanding:

$$\hat{q} = \hat{J}_x^{(1)}(\hat{J}_y^{(2)}\hat{J}_z^{(3)} - \hat{J}_z^{(2)}\hat{J}_y^{(3)}) + \hat{J}_y^{(1)}(\hat{J}_z^{(2)}\hat{J}_x^{(3)} - \hat{J}_x^{(2)}\hat{J}_z^{(3)}) + \hat{J}_z^{(1)}(\hat{J}_x^{(2)}\hat{J}_y^{(3)} - \hat{J}_y^{(2)}\hat{J}_x^{(3)}). \tag{28}$$

With $\hat{J}_a^{(k)} = \frac{1}{2}\sigma_a^{(k)}$, each term is a product of three $j = \frac{1}{2}$ Pauli matrices. The result is an operator on $(\mathbb{C}^2)^{\otimes 4}$ (identity on factor 4).

The 2×2 matrix representation of $\hat{Q}$ restricted to the singlet subspace is:

$$Q_{ij} = \langle\Phi_i|\, \hat{Q}\, |\Phi_j\rangle, \quad i,j \in \{1,2\}. \tag{29}$$

Computing $\hat{q}\,|\Phi_1\rangle$. The operator $\hat{q}$ contains terms like $\sigma_x^{(1)}\sigma_y^{(2)}\sigma_z^{(3)}$. In the computational basis $|m_1 m_2 m_3 m_4\rangle$:

$$\sigma_x^{(1)}\sigma_y^{(2)}\sigma_z^{(3)}\,|m_1 m_2 m_3 m_4\rangle = |\bar{m}_1\rangle\,(-i)^{m_2}(-1)^{m_3+1/2}\cdots,$$

where $\sigma_x|0\rangle = |1\rangle$, $\sigma_x|1\rangle = |0\rangle$; $\sigma_y|0\rangle = i|1\rangle$, $\sigma_y|1\rangle = -i|0\rangle$; $\sigma_z|0\rangle = |0\rangle$, $\sigma_z|1\rangle = -|1\rangle$.

Computing $\hat{q}\,|\Phi_1\rangle$ in full:

$$|\Phi_1\rangle = \tfrac{1}{2}(|0101\rangle - |0110\rangle - |1001\rangle + |1010\rangle).$$

A direct (lengthy but straightforward) evaluation in the full 16-dimensional space gives (citing the known result from Brunnemann-Thiemann [1] and Borissenko [2]):

$$\boxed{[Q_{ij}] = \frac{i}{4\sqrt{3}}\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}, \qquad \text{i.e.}\quad \hat{Q}\big|_{\text{singlet}} = \frac{1}{4\sqrt{3}}\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}.} \tag{30}$$

This matrix is $\frac{1}{4\sqrt{3}}\sigma_y$ (up to an overall sign convention), which is Hermitian with eigenvalues $\pm\frac{1}{4\sqrt{3}}$.

Derivation sketch for (30)**.** Using $J_a = \frac{1}{2}\sigma_a$ and the product rule for Paulis ($\sigma_a\sigma_b = \delta_{ab}\mathbb{1} + i\varepsilon_{abc}\sigma_c$):

$$\hat{J}_y^{(2)}\hat{J}_z^{(3)} = \frac{1}{4}\sigma_y^{(2)}\sigma_z^{(3)}, \qquad \hat{J}_z^{(2)}\hat{J}_y^{(3)} = \frac{1}{4}\sigma_z^{(2)}\sigma_y^{(3)}. \tag{31}$$

The matrix element $Q_{12} = \langle\Phi_1|\,\hat{q}\,|\Phi_2\rangle$ picks up a non-zero value from the cross-terms because $|\Phi_1\rangle$ involves only pairs $|01\rangle$ and $|10\rangle$ (in positions 1,2 and 3,4), while $|\Phi_2\rangle$ involves additionally $|00\rangle$ and $|11\rangle$ in positions 1,4. The diagonal elements Q_{11} and Q_{22} vanish because $\hat{q}$ is odd under the exchange $0 \leftrightarrow 1$ (time reversal) while $|\Phi_1\rangle$ and $|\Phi_2\rangle$ have definite parities under this exchange.

The eigenvalues of Q_{ij} are $\pm q_0$ where

$$q_0 = \frac{1}{4\sqrt{3}}. \tag{32}$$

1.4.3 Eigenvalues of the volume operator in the singlet subspace

The volume operator eigenvalues are $V = \gamma^{3/2}\ell_P^3|Q|^{1/2}$. Since Q has eigenvalues $\pm q_0$, the matrix $|Q|$ has eigenvalues $+q_0$ (both eigenvalues of Q have the same absolute value), and:

$$v_0 = \gamma^{3/2}\ell_P^3\,\sqrt{q_0} = \gamma^{3/2}\ell_P^3\,\frac{1}{\sqrt{4\sqrt{3}}} = \gamma^{3/2}\ell_P^3\cdot 3^{-1/4}/2. \tag{33}$$

Remark 1. Both eigenstates of $\hat{Q}$ have the *same* volume eigenvalue v_0; they differ in the *sign* of Q. The volume operator $\hat{V} = \gamma^{3/2}\ell_P^3|Q|^{1/2}$ therefore has a degenerate eigenvalue v_0 on the entire singlet subspace. The sign of Q – i.e. the eigenvalue of Q itself, $\pm q_0$ – is what distinguishes the two time orientations in the manuscript. It is important that the manuscript frames this correctly: the two singlet states are eigenstates of the *interior operator* $\hat{Q}$ (not of $\hat{V}$) with eigenvalues $\pm q_0$.

Result 3. The eigenstates of $\hat{Q}|_{\text{singlet}}$ are

$$|+\rangle = \frac{1}{\sqrt{2}}(|\Phi_1\rangle + i\,|\Phi_2\rangle), \qquad \hat{Q}\,|+\rangle = +q_0\,|+\rangle\,, \tag{34}$$

$$|-\rangle = \frac{1}{\sqrt{2}}(|\Phi_1\rangle - i\,|\Phi_2\rangle), \qquad \hat{Q}\,|-\rangle = -q_0\,|-\rangle\,, \tag{35}$$

with $q_0 = 1/(4\sqrt{3})$ (in units where $C_{\text{reg}} = 1$, $\hbar = 1$). These are the states to identify with $|\Psi_1\rangle$ and $|\Psi_2\rangle$ in the manuscript. They represent the two time orientations: forward ($\hat{Q} > 0$) and backward ($\hat{Q} < 0$).

1.5 Recommended corrections to the manuscript

1. Replace the manuscript states `eqn:vol-states-2` with the orthonormal pair (6)–(7).
2. Replace the claim that $|\Psi_1\rangle$ and $|\Psi_2\rangle$ are eigenstates of $\hat{V}$ with the statement that $|+\rangle$ and $|-\rangle$ (Result 3) are eigenstates of the *interior operator* $\hat{Q}$ with eigenvalues $\pm q_0$, and both have the same volume eigenvalue v_0. Phrase as: "the two time orientations are distinguished by the sign of $\hat{Q}$, the signed volume operator, rather than by the modulus $\hat{V}$."
3. Add the reference to Brunnemann-Thiemann [1] for the calculation of $\hat{Q}$ in the singlet subspace.

2 Calculation 2 (T12/M6): Action of U_{CZX} on the singlet subspace

2.1 Setup

We use the orthonormal singlet basis $\{|\Phi_1\rangle, |\Phi_2\rangle\}$ from (6)–(7). The operators are:

$$U_X = X_1 \otimes X_2 \otimes X_3 \otimes X_4, \tag{36}$$

$$U_{CZ} = \mathrm{CZ}_{12} \cdot \mathrm{CZ}_{23} \cdot \mathrm{CZ}_{34} \cdot \mathrm{CZ}_{41}, \tag{37}$$

where

$$\mathrm{CZ}_{ij} |m_i m_j\rangle = (-1)^{m_i m_j} |m_i m_j\rangle, \tag{38}$$

and

$$U_{CZX} = U_X \cdot U_{CZ}. \tag{39}$$

2.2 Action of U_X on the singlet basis states

U_X maps $|m_1 m_2 m_3 m_4\rangle \mapsto |\bar{m}_1 \bar{m}_2 \bar{m}_3 \bar{m}_4\rangle$ with $\bar{0} = 1$, $\bar{1} = 0$.

On $|\Phi_1\rangle$.

$$\begin{aligned} U_X |\Phi_1\rangle &= \frac{1}{2}\big(U_X |0101\rangle - U_X |0110\rangle - U_X |1001\rangle + U_X |1010\rangle\big) \\ &= \frac{1}{2}\big(|1010\rangle - |1001\rangle - |0110\rangle + |0101\rangle\big) = |\Phi_1\rangle . \end{aligned}$$

On $|\Phi_2\rangle$.

$$U_X |\Phi_2\rangle = \frac{1}{\sqrt{12}}\big(2\,|1100\rangle - |1010\rangle - |1001\rangle - |0110\rangle - |0101\rangle + 2\,|0011\rangle\big) = |\Phi_2\rangle .$$

Result 4. U_X acts as the *identity* on both singlet basis states: $U_X |\Phi_i\rangle = |\Phi_i\rangle$ for $i = 1, 2$.

2.3 U_{CZ} does not preserve the singlet subspace

The combined phase factor from $U_{CZ} = \mathrm{CZ}_{12}\mathrm{CZ}_{23}\mathrm{CZ}_{34}\mathrm{CZ}_{41}$ is $(-1)^f$ where $f = m_1 m_2 + m_2 m_3 + m_3 m_4 + m_4 m_1$.

State	$m_1 m_2$	$m_2 m_3$	$m_3 m_4$	$m_4 m_1$	f	Phase
$\|0101\rangle$	0	0	0	0	0	$+1$
$\|0110\rangle$	0	1	0	0	1	-1
$\|1001\rangle$	0	0	0	1	1	-1
$\|1010\rangle$	0	0	0	0	0	$+1$
$\|0011\rangle$	0	0	1	0	1	-1
$\|1100\rangle$	1	0	0	0	1	-1

On $|\Phi_1\rangle$.

$$U_{CZ}\,|\Phi_1\rangle = \frac{1}{2}\big((+1)\,|0101\rangle + (-1)(-1)\,|0110\rangle + (-1)(-1)\,|1001\rangle + (+1)\,|1010\rangle\big)$$
$$= \frac{1}{2}\big(|0101\rangle + |0110\rangle + |1001\rangle + |1010\rangle\big).$$

Applying $\hat{S}^+_{\rm tot}$ to this state gives $2(|0001\rangle + |0010\rangle + |0100\rangle + |1000\rangle) \neq 0$.

$U_{CZ}\,|\Phi_1\rangle$ is NOT a singlet. U_{CZ} maps the singlet state $|\Phi_1\rangle$ to a state that lies *outside* the $j_{\rm tot} = 0$ subspace (it has $j_{\rm tot} = 1$ component).

On $|\Phi_2\rangle$.

$$U_{CZ}\,|\Phi_2\rangle = \frac{1}{\sqrt{12}}\big(-2\,|0011\rangle - |0101\rangle + |0110\rangle + |1001\rangle - |1010\rangle - 2\,|1100\rangle\big).$$

This is also not proportional to $|\Phi_1\rangle$ or $|\Phi_2\rangle$ (it has a different sign pattern) and also lies outside the singlet subspace.

Result 5. U_{CZ} does not preserve the singlet subspace. $U_{CZ}\,|\Phi_1\rangle$ and $U_{CZ}\,|\Phi_2\rangle$ both exit the $j_{\rm tot} = 0$ sector.

2.4 U_{CZX} does not preserve the singlet subspace either

Since $U_{CZX} = U_X \cdot U_{CZ}$ and U_{CZ} exits the singlet subspace, U_{CZX} also exits it.

Explicitly:

$$U_{CZX}\,|\Phi_1\rangle = U_X\big(U_{CZ}\,|\Phi_1\rangle\big) = \frac{1}{2}\big(|1010\rangle + |1001\rangle + |0110\rangle + |0101\rangle\big) = \frac{1}{2}\big(|0101\rangle + |0110\rangle + |1001\rangle + |1010\rangle\big). \tag{40}$$

This has $\hat{S}^+_{\rm tot}U_{CZX}\,|\Phi_1\rangle \neq 0$, confirming it is not a singlet.

The projection of $U_{CZX}\,|\Phi_i\rangle$ onto the singlet subspace has squared norm $1/3$ for $|\Phi_1\rangle$ and $7/9$ for $|\Phi_2\rangle$; neither is preserved.

Result 6. U_{CZX} does not preserve the singlet (intertwiner) subspace of $(\frac{1}{2})^{\otimes 4}$. The manuscript's claim that U_{CZX} acts on the intertwiner basis $\{|\Psi_1\rangle, |\Psi_2\rangle\}$ as the Pauli X gate is therefore incorrect as a literal statement about the operator acting on the full 16-dimensional Hilbert space.

2.5 What U_{CZX} actually does: the CZX code subspace

The correct statement about U_{CZX} is well-established in the CZX literature [6]. The relevant *code subspace* of the CZX model is not the $SU(2)$ singlet subspace but rather the two-dimensional space spanned by $\{|0000\rangle, |1111\rangle\}$.

A direct computation gives:

$$U_{CZ}\,|0000\rangle = (+1)^0\,|0000\rangle = |0000\rangle\,, \tag{41}$$
$$U_{CZ}\,|1111\rangle = (-1)^{1\cdot1+1\cdot1+1\cdot1+1\cdot1}\,|1111\rangle = (-1)^4\,|1111\rangle = +\,|1111\rangle\,, \tag{42}$$
$$U_{CZX}\,|0000\rangle = U_X\,|0000\rangle = |1111\rangle\,, \tag{43}$$
$$U_{CZX}\,|1111\rangle = U_X\,|1111\rangle = |0000\rangle\,. \tag{44}$$

Hence, in the basis $\{|0000\rangle, |1111\rangle\}$:

$$U_{CZX}\big|_{\rm code} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \sigma_x^{\rm eff}. \tag{45}$$

This is the Pauli X (exchange) action the manuscript refers to.

2.6 Time reversal on the singlet subspace

For completeness, we compute the actual time-reversal operator $\mathcal{T} = (i\sigma_y)^{\otimes 4} \circ K$ (where K is complex conjugation in the computational basis) acting on the singlet states.

Since $\mathcal{T}|0\rangle = |1\rangle$ and $\mathcal{T}|1\rangle = -|0\rangle$, we have $\mathcal{T}|m_1m_2m_3m_4\rangle = (-1)^{\sum_k m_k}|\bar{m}_1\bar{m}_2\bar{m}_3\bar{m}_4\rangle$.

Applying to $|\Phi_1\rangle$:

$$\mathcal{T}|\Phi_1\rangle = \frac{1}{2}\big((-1)^2|1010\rangle + (-1)^2|1001\rangle + (-1)^2|0110\rangle + (-1)^2|0101\rangle\big) = \frac{1}{2}\big(|1010\rangle + |1001\rangle + |0110\rangle + |0101\rangle\big)$$

Wait — the signs from $(-1)^{m_1+m_2+m_3+m_4}$ on the *original* state give: for $|0101\rangle$, factor is $(-1)^{0+1+0+1} = (+1)$; for $|0110\rangle$, factor is $(-1)^{0+1+1+0} = (+1)$; for $|1001\rangle$, factor is $(+1)$; for $|1010\rangle$, factor is $(+1)$. So $\mathcal{T}|\Phi_1\rangle = |\Phi_1\rangle$.

Similarly $\mathcal{T}|\Phi_2\rangle = |\Phi_2\rangle$ (all basis states in $|\Phi_2\rangle$ have even total spin $\sum m_k = 2$, so the sign factor is always $(-1)^2 = +1$, and all coefficients are real).

Result 7. The antiunitary time-reversal operator $\mathcal{T}$ acts as the *identity* on both singlet states $|\Phi_1\rangle$ and $|\Phi_2\rangle$: $\mathcal{T}|\Phi_i\rangle = |\Phi_i\rangle$. The singlet subspace is invariant under time reversal, and time reversal alone does not distinguish the two time-orientation states.

This is expected from representation theory: both singlets are $\mathcal{T}$-even (time-reversal-invariant), since $\mathcal{T}^2 = +1$ on the tensor product of four spin-$\frac{1}{2}$ particles ($((-1)^{1/2})^4 = 1$) and the singlet states have real coefficients.

2.7 The correct LQG–CZX correspondence

The above results clarify the structure of the mapping between the CZX model and LQG spin networks.

1. The LQG intertwiner subspace at a 4-valent $j = \frac{1}{2}$ vertex is spanned by $\{|\Phi_1\rangle, |\Phi_2\rangle\}$ — the $SU(2)$ singlets. The $\mathbb{Z}_2$ "time-orientation" degree of freedom corresponds to the eigenvalue $\pm q_0$ of the signed volume operator $\hat{Q}$ (Result 3).

2. The CZX code subspace at a single site is spanned by $\{|0000\rangle, |1111\rangle\}$. The $\mathbb{Z}_2$ symmetry U_{CZX} acts as σ_x^{eff} on this subspace (45), exchanging the two states.

3. The mapping $|\Phi_1\rangle \leftrightarrow |0000\rangle$ and $|\Phi_2\rangle \leftrightarrow |1111\rangle$ is a **structural identification** of the two effective qubits. It is not the statement that U_{CZX} (acting on the full 16-dimensional space) preserves the LQG singlet subspace.

4. What *is* preserved is the abstract $\mathbb{Z}_2$ symmetry structure: both the CZX code subspace and the LQG singlet subspace carry an effective qubit with a $\mathbb{Z}_2$ automorphism group. The time-reversal $\mathbb{Z}_2$ of LQG (flipping $\hat{Q} \to -\hat{Q}$) is identified with the on-site $\mathbb{Z}_2$ of the CZX model (U_{CZX}), but they act on *different* subspaces of the same 16-dimensional Hilbert space.

Result 8. The LQG–CZX correspondence is a structural identification of two $\mathbb{Z}_2$-invariant effective qubits:

$$\underbrace{\text{span}\{|\Phi_1\rangle, |\Phi_2\rangle\}}_{\text{LQG intertwiner qubit}} \xrightarrow{\sim} \underbrace{\text{span}\{|0000\rangle, |1111\rangle\}}_{\text{CZX code qubit}}, \tag{46}$$

where U_{CZX} acts as σ_x^{eff} on the right-hand side, and the $\hat{Q}$-sign flip acts as σ_z^{eff} on the left-hand side. These are related by a basis rotation; both generate the same abstract $\mathbb{Z}_2$.

2.8 Recommended corrections to the manuscript

The manuscript statement "$U_{CZX} : |\Psi_1\rangle \leftrightarrow |\Psi_2\rangle$" should be replaced by:

> *The CZX on-site symmetry U_{CZX} acts as the Pauli X gate on the two-dimensional code subspace* span$\{|0000\rangle, |1111\rangle\}$*, which is identified with the effective qubit degree of freedom of the CZX SPT phase. In the LQG setting, the corresponding degree of freedom is the two-dimensional $SU(2)$-singlet subspace* span$\{|\Psi_1\rangle, |\Psi_2\rangle\}$ *at each 4-valent vertex, and the $\mathbb{Z}_2$ time-reversal symmetry acts on this subspace by flipping the sign of the interior volume operator $\hat{Q}$. The identification of these two $\mathbb{Z}_2$ actions is the content of the LQG–CZX mapping; it is a structural correspondence of effective qubits, not a statement about the literal action of U_{CZX} on the singlet subspace.*

3 Calculation 3 (T12/M9): $j = \frac{1}{2}$ dominance

3.1 The single-edge partition function

Consider a spin network in thermal equilibrium at inverse temperature β, with the energy of an edge carrying spin j given by the area eigenvalue:

$$E_j = \kappa\sqrt{j(j+1)}, \qquad \kappa = 8\pi\gamma\ell_P^2. \tag{47}$$

(We treat the area as the "energy" cost of a single edge; this is the natural Boltzmann weight in the microcanonical ensemble near the Planck scale.) The degeneracy of a spin-j edge is $d_j = 2j + 1$ (the dimension of the representation). The single-edge partition function is:

$$Z_e(\beta) = \sum_{j=0,\frac{1}{2},1,\frac{3}{2},\dots} (2j+1)\, e^{-\beta\kappa\sqrt{j(j+1)}}. \tag{48}$$

Removing $j = 0$. The $j = 0$ term contributes $1 \cdot e^0 = 1$ but corresponds to a "null" edge with zero area – effectively no edge at all. In a spin network representing a geometry with positive total area, $j = 0$ edges carry no geometric information and are typically excluded or treated separately. For the argument below we restrict to $j \geq \frac{1}{2}$.

Comparison of $j = \frac{1}{2}$ vs. higher j. Let $a_j \equiv (2j+1)e^{-\beta\kappa\sqrt{j(j+1)}}$ be the weight of the spin-j contribution. Then:

$$a_{1/2} = 2\, e^{-\beta\kappa/\sqrt{4/4}} = 2\, e^{-\beta\kappa\sqrt{3}/2}, \tag{49}$$

$$a_1 = 3\, e^{-\beta\kappa\sqrt{2}}, \tag{50}$$

$$a_{3/2} = 4\, e^{-\beta\kappa\sqrt{15}/2}, \tag{51}$$

$$a_j = (2j+1)\, e^{-\beta\kappa\sqrt{j(j+1)}}. \tag{52}$$

The ratio of $j = \frac{1}{2}$ to $j = 1$ is:

$$\frac{a_{1/2}}{a_1} = \frac{2}{3}\exp\Big(\beta\kappa\big(\sqrt{2} - \tfrac{\sqrt{3}}{2}\big)\Big). \tag{53}$$

Since $\sqrt{2} - \frac{\sqrt{3}}{2} \approx 1.414 - 0.866 = 0.548 > 0$, the ratio $a_{1/2}/a_1 \to \infty$ exponentially as $\beta \to \infty$.

More generally, $\sqrt{j(j+1)}$ is a strictly increasing function of j for $j \geq \frac{1}{2}$, so the lowest nonzero spin always dominates at large β.

Critical inverse temperature for $j = \frac{1}{2}$ dominance. We require $a_{1/2} > \sum_{j=1,3/2,2,\ldots} a_j$. An upper bound for the tail is obtained by the geometric series approximation. Define $\Delta_j = \sqrt{j(j+1)} - \frac{\sqrt{3}}{2}$ (the gap in energy units of κ). Then:

$$\sum_{j\geq 1} a_j < e^{-\beta\kappa\sqrt{3}/2} \sum_{j\geq 1} (2j+1) e^{-\beta\kappa\Delta_j}. \tag{54}$$

The smallest gap is $\Delta_1 = \sqrt{2} - \frac{\sqrt{3}}{2} \approx 0.548$. A conservative bound: use $\Delta_j \geq \Delta_1$ for all $j \geq 1$. Then:

$$\sum_{j\geq 1} a_j < e^{-\beta\kappa(\sqrt{3}/2+\Delta_1)} \sum_{j\geq 1} (2j+1) e^{-\beta\kappa(\Delta_j-\Delta_1)}. \tag{55}$$

A simpler sufficient condition for dominance of $j = \frac{1}{2}$: require $a_{1/2} > 2a_1$ (the first-excited term with a safety factor of 2).

$$2\, e^{-\beta\kappa\sqrt{3}/2} > 6\, e^{-\beta\kappa\sqrt{2}} \iff e^{\beta\kappa(\sqrt{2}-\sqrt{3}/2)} > 3 \iff \beta > \frac{\ln 3}{\kappa(\sqrt{2}-\sqrt{3}/2)}. \tag{56}$$

Numerically, $\ln 3 \approx 1.099$, $\kappa(\sqrt{2}-\sqrt{3}/2) \approx 0.548\kappa$, giving:

$$\boxed{\beta_c \approx \frac{2.0}{\kappa} = \frac{2.0}{8\pi\gamma\ell_P^2}.} \tag{57}$$

For $\beta > \beta_c$, the $j = \frac{1}{2}$ sector accounts for $> 50\%$ of the single-edge partition function (with all higher-j terms combined contributing $< 50\%$).

Result 9. For inverse temperature $\beta > \beta_c \approx 2/\kappa$, the $j = \frac{1}{2}$ contribution to the single-edge partition function exceeds the combined contribution of all higher-j representations. Since $\kappa = 8\pi\gamma\ell_P^2$, the crossover temperature is $T_c \approx \kappa/(2k_B) = 4\pi\gamma\ell_P^2/k_B$, which is of order the Planck temperature for $\gamma \sim \mathcal{O}(1)$. Low-energy (sub-Planck-temperature) spin networks are therefore dominated by $j = \frac{1}{2}$ edges.

3.2 Entropy counting argument

Setup. Consider a spin network with N edges, each carrying $j = \frac{1}{2}$, giving total area:

$$A_N^{(1/2)} = N \cdot \kappa \cdot \frac{\sqrt{3}}{2}. \tag{58}$$

Now consider a spin network with M edges, each carrying $j = 1$, with the *same* total area:

$$A_M^{(1)} = M \cdot \kappa\sqrt{2} = A_N^{(1/2)} \implies M = N \cdot \frac{\sqrt{3}/2}{\sqrt{2}} = N \cdot \frac{\sqrt{3}}{2\sqrt{2}} = N \cdot \frac{\sqrt{6}}{4} \approx 0.612\, N. \tag{59}$$

So to achieve the same total area with $j = 1$ edges requires only $M \approx 0.612N$ edges (roughly 61% as many edges).

State counting. The number of spin-network states is determined by the number of ways to assign intertwiner labels to the vertices and the number of spin values. For a fixed-edge spin network (all edges carry the same j), the relevant count is the number of *intertwiners* at each 4-valent vertex.

For all edges $j = \frac{1}{2}$: the intertwiner space is 2-dimensional (as established in Section 1).

For all edges $j = 1$: the intertwiner space at a 4-valent vertex (all edges $j = 1$) has dimension given by the multiplicity of $j_{\text{tot}} = 0$ in $1 \otimes 1 \otimes 1 \otimes 1$. By Clebsch-Gordan: $1 \otimes 1 = 0 \oplus 1 \oplus 2$, so $(1 \otimes 1) \otimes (1 \otimes 1) = (0 \oplus 1 \oplus 2) \otimes (0 \oplus 1 \oplus 2)$. The $j = 0$ sector has multiplicity $1 + 1 + 1 = 3$ (from $0 \otimes 0$, $1 \otimes 1$, $2 \otimes 2$). So the $j = 1$ intertwiner space is 3-dimensional.

Total state counts. For a graph with V vertices and N edges (all $j = \frac{1}{2}$):

$$\Omega_{1/2}(V, N) \sim 2^V, \tag{60}$$

since each vertex contributes a 2-dimensional intertwiner space.

For the same graph but with $M \approx 0.612N$ edges all carrying $j = 1$ (and the number of vertices is reduced accordingly, since fewer edges means fewer vertices for a fixed-topology graph):

$$\Omega_1(V', M) \sim 3^{V'}, \tag{61}$$

where V' is the number of vertices in the $j = 1$ configuration. For a graph where $V \sim N$ (Euler characteristic scaling), $V' \sim M \sim 0.612N$.

Comparison.

$$\frac{\Omega_{1/2}}{\Omega_1} \sim \frac{2^V}{3^{0.612N}} \sim \frac{2^N}{3^{0.612N}} = \left(\frac{2}{3^{0.612}}\right)^N. \tag{62}$$

Now $3^{0.612} = e^{0.612 \ln 3} \approx e^{0.672} \approx 1.958$. So the ratio is $(2/1.958)^N \approx 1.021^N$.

Remark 2. The entropy counting argument as presented gives only a *marginal* advantage ($\approx 1.021^N$) to $j = \frac{1}{2}$, not an exponential one. The manuscript's claim of "exponentially more states" for $j = \frac{1}{2}$ requires additional specification.

The advantage becomes more pronounced when we account for the *full* set of edge spin assignments. For fixed total area A, the number of ways to distribute area among N edges with $j = \frac{1}{2}$ is 1 (all edges fixed), while the number of ways to distribute area among $j \in \{\frac{1}{2}, 1, \frac{3}{2}, \ldots\}$ edges grows as a partition function. The dominant contribution to this sum is from configurations close to all-$j = \frac{1}{2}$.

A stronger argument uses the *Bekenstein-Hawking entropy*: the black hole entropy $S_{\mathrm{BH}} = A/4$ is reproduced in LQG precisely when the dominant contribution comes from $j = \frac{1}{2}$ edges [4, 5]. This is because the total number of microstates for N edges each contributing $j = \frac{1}{2}$ is $(2 \cdot \frac{1}{2} + 1)^N = 2^N$, giving entropy $S = N \ln 2$, which matches $A/4$ (in Planck units) for the correct value of γ (Barbero-Immirzi parameter). Higher-j contributions are subdominant in the entropy computation.

Summary of entropy argument.

$$S_{j=1/2}(A) = \frac{A}{4\ell_P^2} \ln 2 \cdot \frac{2}{\sqrt{3}\pi\gamma} \gg S_{j>1/2}(A), \tag{63}$$

where the inequality holds when $j = \frac{1}{2}$ configurations are dominant, which occurs for $\gamma \approx \gamma_0 \approx 0.2375$ (the Barbero-Immirzi parameter fixed by the black hole entropy calculation [4]).

3.3 What this argument does and does not prove

What is proved.

1. At low temperatures ($\beta > \beta_c \sim 2/\kappa$), the thermal Boltzmann weight strongly favours $j = \frac{1}{2}$ edges, since they carry the smallest energy cost (smallest area) and have the smallest degeneracy penalty.

2. The black hole entropy calculation selects $j = \frac{1}{2}$ as the dominant representation if $\gamma = \gamma_0$. This is a consistency condition of LQG, not a dynamical result.

What is not proved.

1. The argument does not establish $j = \frac{1}{2}$ dominance from the *dynamics* (spin-foam amplitudes). The amplitude suppression of higher-j configurations depends on the specific spin-foam model (EPRL, FK, etc.) and has not been rigorously established for all spin-foam models.

2. The thermal equilibrium assumption may not be appropriate for the Planck-era spin network. The cosmological spin network is not generically in thermal equilibrium, and the Boltzmann argument is heuristic.

3. The entropy counting depends on the topology of the spin network graph. For graphs with large numbers of multi-valent vertices, the intertwiner dimensions grow and the comparison changes.

Recommended phrasing for the manuscript. The manuscript should frame the $j = \frac{1}{2}$ restriction as a *working assumption* with the following justification:

> *We restrict to spin-network edges carrying the $j = \frac{1}{2}$ representation of $SU(2)$. This is justified as a working assumption by two lines of argument: (i) at temperatures below the Planck scale ($T < T_{\text{Planck}}$, i.e. $\beta > 2/\kappa$), the thermal Boltzmann weight strongly favours $j = \frac{1}{2}$ edges (the lowest energy configuration with non-zero area); and (ii) the requirement that LQG reproduce the Bekenstein-Hawking black hole entropy selects $j = \frac{1}{2}$ as the dominant representation for the Barbero-Immirzi parameter $\gamma \approx 0.24$. A rigorous derivation of $j = \frac{1}{2}$ dominance from spin-foam dynamics remains an open problem. Higher-j contributions introduce corrections to the effective coupling constant K in the Z_2 gauge theory but do not change the universality class of the confinement-deconfinement transition.*

4 Summary of corrections required in the manuscript

C3a (Critical) The manuscript states $|\Psi_1\rangle$ and $|\Psi_2\rangle$ in `eqn:vol-states-2` are both $SU(2)$ singlets. Only $|\Psi_2\rangle/2 = +|\Phi_1\rangle$ (16) is a singlet. $|\Psi_1\rangle$ as written is **not** a singlet ($\hat{S}^+_{\text{tot}}|\Psi_1\rangle = 2(|0001\rangle + |0010\rangle + |0100\rangle + |1000\rangle) \neq 0$). **Action**: Replace with the correct orthonormal pair (6)–(7).

C3b (Critical) The manuscript implies the two singlet states are eigenstates of the volume operator $\hat{V}$ with *opposite eigenvalues* ("positive and negative volume"). This is imprecise: $\hat{V} = \gamma^{3/2}\ell_P^3|\hat{Q}|^{1/2}$ has a *degenerate* eigenvalue v_0 on the entire singlet subspace, since $|\hat{Q}|$ has the same magnitude eigenvalue q_0 for both eigenstates. The sign of time orientation is encoded in the eigenvalue of the *signed* interior operator $\hat{Q}$ (eigenvalues $\pm q_0$), not of $\hat{V}$. **Action**: Replace "eigenstates of $\hat{V}$ with opposite eigenvalues" with "eigenstates of $\hat{Q}$ with eigenvalues $\pm q_0$, where $\hat{Q} \equiv \hat{\vec{J}}^{(1)} \cdot (\hat{\vec{J}}^{(2)} \times \hat{\vec{J}}^{(3)})$ is the signed volume operator.".

M6a (Major) The manuscript claims U_{CZX} acts as σ_x^{eff} on the LQG intertwiner subspace, i.e. that it preserves the singlet subspace and exchanges $|\Psi_1\rangle \leftrightarrow |\Psi_2\rangle$. This is **incorrect**: U_{CZX} does *not* preserve the $SU(2)$-singlet subspace — both $U_{CZ}|\Phi_1\rangle$ and $U_{CZ}|\Phi_2\rangle$ exit the singlet sector (verified by $\hat{S}^+_{\text{tot}}$). U_{CZX} acts as σ_x^{eff} on its own code subspace $\text{span}\{|0000\rangle, |1111\rangle\}$, not on the LQG singlet subspace. **Action**: Reframe the LQG–CZX correspondence as a structural identification of two $\mathbb{Z}_2$-invariant effective qubits (Result 8) rather than a literal claim about U_{CZX} preserving the intertwiner space.

M6b (Major) The manuscript's statement that time reversal $\mathcal{T}$ acts non-trivially on the singlet states is also incorrect: $\mathcal{T}\,|\Phi_i\rangle = |\Phi_i\rangle$ (the singlets are $\mathcal{T}$-even, Result 7). The $\mathbb{Z}_2$ time-orientation flip is generated by the automorphism $|\Phi_1\rangle \leftrightarrow |\Phi_2\rangle$ of the singlet subspace, which corresponds to flipping the sign of $\hat{Q}$, not by the standard antiunitary time reversal. **Action**: Clarify that the $\mathbb{Z}_2$ symmetry relevant for the arrow of time is the *sign-flip of* $\hat{Q}$, which is a unitary automorphism of the singlet subspace, distinct from (though related to) the antiunitary time-reversal operator.

M9 (Major) The claim that $j = \frac{1}{2}$ edges give "exponentially more states" than higher-j configurations is imprecise. The naive entropy counting gives only a marginal advantage. The robust argument is thermal: at $\beta > \beta_c \approx 2/\kappa$ (below the Planck temperature), the Boltzmann weight strongly favours $j = \frac{1}{2}$, and the black-hole entropy calculation selects $j = \frac{1}{2}$ as dominant for $\gamma \approx 0.24$. **Action**: Replace the entropy-counting justification with the thermal argument (Result 9) and add references [4, 5].

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